VIP Validation Report

Deployment: CI Connect 2026.10.0 + WB 2026.10.1+239.pro3 + PM 2026.11.0
Generated: 2026-11-11 20:04:00 UTC

Products Under Test

Product	URL	Version	Results
Connect	`http://localhost:3939`	2026.10.0	9 failed (9 tests)
Workbench	`http://localhost:8787`	2026.10.1+239.pro3	2 failed (2 tests)
Package Manager	`http://localhost:4242`	2026.11.0	5 failed (5 tests)

Summary

Total	25
Passed	0
Failed	25
Skipped	0
Status	**FAIL**

Provenance

VIP version	2026.11.0
Run duration	1.9s
Python	3.14.4
Platform	Linux-7.0.0-1019-azure-x86_64-with-glibc2.43
Mode	full
Exit status	1 (tests were collected and run, but some failed)

Failures & Skips

Every check that did not pass, in full. Passing checks are counted above, and every check appears in Detailed Results below.

Failed (25)

FAIL **User can sign out of Workbench** Workbench
`src/vip_tests/workbench/test_auth.py::test_workbench_signout[chromium]  0.00s`

Test procedure
```
› Given Workbench is accessible and I am logged in
› When I sign out of Workbench
› Then I am redirected to the Workbench login page
```

test_workbench_signout[chromium]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8031f930>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
```

```
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Workbench is accessible and I am logged in', line_number=14,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_workbench_signout[chromium]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- The sign-out button or menu changed location between Workbench versions
- A reverse proxy is caching the post-logout redirect

Suggested next steps:

1. Manually sign out in a browser and confirm you land back on the login page
2. Check any reverse proxy or CDN in front of Workbench for cached redirects

Documentation: https://docs.posit.co/ide/server-pro/authenticating_users/authenticating_users.html

FAIL **Workbench server is reachable** Prerequisites

src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Workbench]
0.00s

Test procedure

```
› Given <product> is configured in vip.toml
› When I request the <product> health endpoint
› Then the server responds with a successful status code
```

test_product_server_is_reachable[Workbench]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017c6e0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Workbench is configured in vip.toml', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_product_server_is_reachable[Workbench]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

FAIL **Package Manager server is reachable** Prerequisites

src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Package Manager] 0.00s

Test procedure

› Given <product> is configured in vip.toml
› When I request the <product> health endpoint
› Then the server responds with a successful status code

test_product_server_is_reachable[Package Manager]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017f310>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is configured in vip.toml', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_product_server_is_reachable[Package Manager]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

FAIL **This check intentionally fails to demonstrate failure rendering** Prerequisites

As someone evaluating a VIP report

`src/vip_tests/prerequisites/test_expected_failure.py::test_intentional_failure  0.00s`

Test procedure

› Given a check that is written to fail on purpose
› When the check runs as part of this example report
› Then it fails by design, not because anything is actually broken

test_intentional_failure: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017fa10>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='a check that is written to fail on purpose', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_intentional_failure>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- This check is designed to always fail, so the example report has a failure card to show
- No product or configuration problem is involved

Suggested next steps:

1. No action needed. This failure is intentional and does not indicate a real problem
2. It only runs when VIP_ENABLE_EXPECTED_FAILURE_DEMO=1 is set

Documentation: https://github.com/posit-dev/vip/issues/73

FAIL **User can log in to Workbench via the web UI** Workbench

`src/vip_tests/workbench/test_auth.py::test_workbench_login[chromium]  0.00s`

Test procedure

› Given Workbench is accessible at the configured URL
› When a user navigates to the Workbench login page and enters valid credentials
› Then the Workbench homepage is displayed
› And the current user element is visible and non-empty in the header

test_workbench_login[chromium]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c4da270>
```

```
    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Workbench is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_workbench_login[chromium]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Incorrect test credentials in vip.toml or VIP_TEST_USERNAME/VIP_TEST_PASSWORD
- Auth provider misconfiguration (SAML, OIDC, LDAP, PAM)
- Workbench login page markup changed or is unreachable

Suggested next steps:

1. Verify credentials work by logging in manually at the Workbench URL
2. Check the auth provider configuration in rserver.conf / auth-pam or auth-saml settings
3. If using an external IdP, try `--interactive-auth` or `--headless-auth` instead of a password test

Documentation: https://docs.posit.co/ide/server-pro/authenticating_users/authenticating_users.html

FAIL **Connect version matches configuration** Prerequisites

src/vip_tests/prerequisites/test_versions.py::test_connect_version 0.00s

Test procedure

› Given Connect is configured in vip.toml with a version expectation
› When I fetch the Connect server version
› Then the Connect version matches the configured value

test_connect_version: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017dc50>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
```

```
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml with a version expectation',
    line_number=8, indent=4, keyword='Given', docstring=None, datatable=None, failed=False,
    scenario=None, background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_connect_version>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Connect was upgraded but vip.toml's [connect] version was not updated
- vip.toml points at the wrong Connect deployment

Suggested next steps:

1. Check the running version: `curl -H 'Authorization: Key <key>' <connect-url>/__api__/server_settings`
2. Update the version field under [connect] in vip.toml to match

Documentation: https://docs.posit.co/connect/admin/server-management/

FAIL **Package Manager version matches configuration** Prerequisites

src/vip_tests/prerequisites/test_versions.py::test_package_manager_version 0.00s

Test procedure

```
› Given Package Manager is configured in vip.toml with a version expectation
› When I fetch the Package Manager server version
› Then the Package Manager version matches the configured value
```

test_package_manager_version: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017fe70>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is configured in vip.toml with a version
    expectation', line_number=18, indent=4, keyword='Given', docstring=None, datatable=None,
    failed=False, scenario=None, background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_package_manager_version>
```

```
    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- Package Manager was upgraded but vip.toml's [package_manager] version was not updated
- vip.toml points at the wrong Package Manager deployment

Suggested next steps:
1. Check the running version: `curl <pm-url>/__api__/status`
2. Update the version field under [package_manager] in vip.toml to match

Documentation: https://docs.posit.co/rspm/admin/upgrading/

FAIL **User can log in via the web UI** Connect

src/vip_tests/connect/test_auth.py::test_connect_login_ui 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› When a user navigates to the Connect login page
› And enters valid credentials
› Then the user is successfully authenticated
› And the Connect dashboard is displayed
```

test_connect_login_ui: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017ce50>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_connect_login_ui>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?
```

```
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- Incorrect test credentials
- Auth provider misconfiguration (SAML, OIDC, LDAP)
- Connect login page has changed or is unreachable
- Playwright browser not installed

Suggested next steps:
1. Verify credentials work by logging in manually at the Connect URL
2. Check auth provider settings in Connect admin panel
3. Run `uv run vip install` to ensure the browser and its system libraries are installed
4. If using external IdP, try `--interactive-auth` or `--headless-auth`

Documentation: https://docs.posit.co/connect/admin/authentication/

FAIL **Connect server is reachable** Prerequisites

src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Connect] 0.00s

Test procedure

```
› Given <product> is configured in vip.toml
› When I request the <product> health endpoint
› Then the server responds with a successful status code
```

test_product_server_is_reachable[Connect]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b000>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_product_server_is_reachable[Connect]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

FAIL **API key authentication works** Connect

src/vip_tests/connect/test_auth.py::test_connect_login_api 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› And a valid API key is configured
› When I request the current user via the API
› Then the API returns user information
```

test_connect_login_api: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b2a0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=16,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_connect_login_api>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- VIP_CONNECT_API_KEY not set or invalid
- API key expired or revoked
- Connect API is not accessible

Suggested next steps:

1. Verify the API key: `curl -H 'Authorization: Key <key>' <connect-url>/__api__/v1/user`
2. Generate a new API key from the Connect UI if needed
3. Check that the VIP_CONNECT_API_KEY environment variable is exported

Documentation: https://docs.posit.co/connect/user/api-keys/

FAIL **Admin user exists and has admin privileges** Connect

src/vip_tests/connect/test_users.py::test_admin_user 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› When I retrieve the current user profile
› Then the user has admin privileges
```

test_admin_user: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017cb40>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
```

```
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_admin_user>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The configured API key belongs to a publisher or viewer, not an administrator
- The API key's owning user was demoted or deactivated

Suggested next steps:
1. Check the user's role in the Connect admin panel under People
2. Generate a new API key from an administrator account and update vip.toml or VIP_CONNECT_API_KEY

Documentation: https://docs.posit.co/connect/admin/user-management/

FAIL **Groups can be listed** Connect

src/vip_tests/connect/test_users.py::test_list_groups 0.00s

Test procedure

› Given Connect is accessible at the configured URL
› When I list all groups
› Then the response is successful

test_list_groups: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b700>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=19,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_list_groups>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Group management is backed by an external directory (LDAP/SAML) that is unreachable
- The API key does not have administrator privileges

Suggested next steps:

1. Check Connect's authentication provider configuration for group-sync errors
2. Verify the API key belongs to an administrator

Documentation: https://docs.posit.co/connect/admin/user-management/

FAIL **Users can be listed** Connect

src/vip_tests/connect/test_users.py::test_list_users 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› When I list all users
› Then the user list is not empty
› And the test user exists in the user list
```

test_list_users: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017d390>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=13,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_list_users>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
```

```
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The test user named by [auth] username in vip.toml (or VIP_TEST_USERNAME) does not exist on Connect
- Connect has no users other than the bootstrap administrator

Suggested next steps:
1. Create the test user in Connect's admin panel or via `POST /__api__/v1/users`
2. Confirm [auth] username in vip.toml matches an existing Connect username exactly

Documentation: https://docs.posit.co/connect/admin/user-management/

FAIL **Expected R versions are available on Connect** Connect

src/vip_tests/connect/test_runtime_versions.py::test_r_versions 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› And expected R versions are specified in vip.toml
› When I query Connect for available R versions
› Then all expected R versions are present
```

test_r_versions: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40bc40>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_r_versions>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The R version was never installed on the Connect server
- vip.toml's [runtimes] r_versions list is out of date with what was actually installed

Suggested next steps:
1. List installed R versions: `docker exec <container> ls /opt/R` or check the Connect admin panel
2. Install the missing R version, or correct r_versions under [runtimes] in vip.toml

Documentation: https://docs.posit.co/connect/admin/r/

FAIL **Expected Python versions are available on Connect** Connect

src/vip_tests/connect/test_runtime_versions.py::test_python_versions 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› And expected Python versions are specified in vip.toml
› When I query Connect for available Python versions
› Then all expected Python versions are present
```

test_python_versions: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017d7f0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=14,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_python_versions>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The Python version was never installed on the Connect server
- vip.toml's [runtimes] python_versions list is out of date with what was actually installed

Suggested next steps:
1. List installed Python versions: `docker exec <container> ls /opt/python` or check the Connect admin panel
2. Install the missing Python version, or correct python_versions under [runtimes] in vip.toml

Documentation: https://docs.posit.co/connect/admin/python/

FAIL **Expected Quarto versions are available** Connect

src/vip_tests/connect/test_runtime_versions.py::test_quarto_versions 0.00s

Test procedure

› Given Connect is accessible at the configured URL
› When I query Connect for available Quarto versions
› Then at least one Quarto version is available

test_quarto_versions: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c356c80>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=20,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_quarto_versions>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Quarto was never installed on the Connect server
- Quarto is installed but not on Connect's configured search path

Suggested next steps:

1. Check installed Quarto versions in the Connect admin panel under Versions
2. Install Quarto and confirm Connect's Quarto.Executable / search path configuration

Documentation: https://docs.posit.co/connect/admin/quarto/

FAIL **Connect system checks can be run and the report downloaded** Connect

src/vip_tests/connect/test_system_checks.py::test_connect_system_checks 0.00s

Test procedure

› Given Connect is accessible at the configured URL
› And a valid API key is configured
› When I trigger a new system check run via the Connect API
› Then the system check report is returned
› And I can download the system check report artifact

test_connect_system_checks: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017e3c0>

    def __enter__(self):
```

```
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_connect_system_checks>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The API key does not have administrator privileges
- A system check itself is failing on the host (e.g. missing sandbox binary, disk space)
- The system check run timed out before reaching a 'done' status

Suggested next steps:
1. Re-run system checks from the Connect admin panel under System Information
2. Review the individual check results for the specific failing check
3. Verify the API key belongs to an administrator

Documentation: https://docs.posit.co/connect/user/system-checks/

FAIL **Invalid API key returns 401** Security

`src/vip_tests/security/test_error_handling.py::test_invalid_api_key_returns_401  0.00s`

Test procedure

```
› Given Connect is configured in vip.toml
› When I make an API request to Connect with an invalid key
› Then the response status is 401
```

test_invalid_api_key_returns_401: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f805b3930>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
```

```
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=13, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_invalid_api_key_returns_401>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- An upstream proxy or load balancer is granting access before Connect validates the key
- Connect's authentication is misconfigured to accept any Authorization header

Suggested next steps:
1. Reproduce with `curl -i -H 'Authorization: Key invalid' <connect-url>/__api__/v1/content`
2. Check any reverse proxy or gateway in front of Connect for auth bypass rules

Documentation: https://docs.posit.co/connect/admin/authentication/

FAIL **Unauthenticated API request returns 401** Security

src/vip_tests/security/test_error_handling.py::test_unauthenticated_api_request_returns_401
0.00s

Test procedure
```
› Given Connect is configured in vip.toml
› When I make an unauthenticated API request to Connect
› Then the response status is 401
```

test_unauthenticated_api_request_returns_401: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c356f20>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_unauthenticated_api_request_returns_401>
```

```
    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- An upstream proxy or load balancer is granting access before Connect sees the request
- Connect's authentication is misconfigured to allow anonymous access to authenticated endpoints

Suggested next steps:
1. Reproduce with `curl -i <connect-url>/__api__/v1/content` and inspect the status code
2. Check any reverse proxy or gateway in front of Connect for auth bypass rules

Documentation: https://docs.posit.co/connect/admin/authentication/

FAIL **Non-existent endpoint returns 404** Security

src/vip_tests/security/test_error_handling.py::test_nonexistent_endpoint_returns_404 0.00s

Test procedure

```
› Given Connect is configured in vip.toml
› When I request a non-existent endpoint on Connect
› Then the response status is 404
```

test_nonexistent_endpoint_returns_404: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b3f0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=18, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_nonexistent_endpoint_returns_404>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- An upstream proxy or load balancer is intercepting the request before Connect replies
- A reverse proxy is rewriting unmatched paths to a different response

Suggested next steps:
1. Reproduce with `curl -i <connect-url>/__api__/v1/does-not-exist-vip-test`
2. Check reverse proxy path-rewrite and catch-all rules in front of Connect

Documentation: https://docs.posit.co/connect/admin/proxy/

FAIL **CRAN mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_cran_mirror 0.00s

Test procedure

```
› Given Package Manager is running
› When I query the CRAN repository for the "Matrix" package
› Then the package is found in the repository
```

test_cran_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f816b58d0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_cran_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No CRAN-type repository is configured in Package Manager
- The CRAN source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **Bioconductor mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_bioconductor_mirror 0.00s

Test procedure

```
› Given Package Manager is running
› When I query the Bioconductor repository for the "BiocGenerics" package
› Then the package is found in the repository
```

test_bioconductor_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c4da430>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=18, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_bioconductor_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- No Bioconductor-type repository is configured in Package Manager
- The Bioconductor source has not finished its initial sync

Suggested next steps:

1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **OpenVSX mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_openvsx_mirror 0.00s

Test procedure

```
› Given Package Manager is running
› When I query the OpenVSX repository for the "golang.Go" extension
› Then the package is found in the repository
```

test_openvsx_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c4da350>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
```

```
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=23, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_openvsx_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No OpenVSX-type repository is configured in Package Manager
- The OpenVSX source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **PyPI mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_pypi_mirror 0.00s

Test procedure

```
› Given Package Manager is running
› When I query the PyPI repository for the "requests" package
› Then the package is found in the repository
```

test_pypi_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8166c440>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
```

```
step = Step(type='given', name='Package Manager is running', line_number=13, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_pypi_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No PyPI-type repository is configured in Package Manager
- The PyPI source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **At least one repository is configured** Package Manager

src/vip_tests/package_manager/test_repos.py::test_repo_exists 0.00s

Test procedure

› Given Package Manager is running
› When I list all repositories
› Then at least one repository exists

test_repo_exists: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027d8b3ee0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=28, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_repo_exists>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
```

```
                                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- Package Manager was installed but no repositories were ever created
- Repositories exist but were deleted or renamed

Suggested next steps:
1. Create a repository: `rspm create repo --name=<name> --type=<cran|python|bioconductor>`
2. List existing repositories: `rspm list repos`

Documentation: https://docs.posit.co/rspm/admin/repositories/

Detailed Results

Connect
9 tests — 9 failed

FAIL **User can log in via the web UI** Connect
src/vip_tests/connect/test_auth.py::test_connect_login_ui 0.00s

Test procedure
› Given Connect is accessible at the configured URL
› When a user navigates to the Connect login page
› And enters valid credentials
› Then the user is successfully authenticated
› And the Connect dashboard is displayed

test_connect_login_ui: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017ce50>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_connect_login_ui>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- Incorrect test credentials
- Auth provider misconfiguration (SAML, OIDC, LDAP)
- Connect login page has changed or is unreachable
- Playwright browser not installed

Suggested next steps:
1. Verify credentials work by logging in manually at the Connect URL
2. Check auth provider settings in Connect admin panel
3. Run `uv run vip install` to ensure the browser and its system libraries are installed
4. If using external IdP, try `--interactive-auth` or `--headless-auth`

Documentation: https://docs.posit.co/connect/admin/authentication/

FAIL **API key authentication works** Connect
src/vip_tests/connect/test_auth.py::test_connect_login_api 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› And a valid API key is configured
› When I request the current user via the API
› Then the API returns user information
```

test_connect_login_api: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b2a0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=16,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_connect_login_api>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- VIP_CONNECT_API_KEY not set or invalid
- API key expired or revoked
- Connect API is not accessible

Suggested next steps:

1. Verify the API key: `curl -H 'Authorization: Key <key>' <connect-url>/__api__/v1/user`
2. Generate a new API key from the Connect UI if needed
3. Check that the VIP_CONNECT_API_KEY environment variable is exported

Documentation: https://docs.posit.co/connect/user/api-keys/

FAIL **Admin user exists and has admin privileges** Connect

`src/vip_tests/connect/test_users.py::test_admin_user` 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› When I retrieve the current user profile
› Then the user has admin privileges
```

test_admin_user: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017cb40>

    def __enter__(self):
```

```
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_admin_user>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The configured API key belongs to a publisher or viewer, not an administrator
- The API key's owning user was demoted or deactivated

Suggested next steps:
1. Check the user's role in the Connect admin panel under People
2. Generate a new API key from an administrator account and update vip.toml or VIP_CONNECT_API_KEY

Documentation: https://docs.posit.co/connect/admin/user-management/

FAIL **Groups can be listed** Connect

src/vip_tests/connect/test_users.py::test_list_groups 0.00s

Test procedure

› Given Connect is accessible at the configured URL
› When I list all groups
› Then the response is successful

test_list_groups: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b700>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
```

```
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=19,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_list_groups>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- Group management is backed by an external directory (LDAP/SAML) that is unreachable
- The API key does not have administrator privileges

Suggested next steps:
1. Check Connect's authentication provider configuration for group-sync errors
2. Verify the API key belongs to an administrator

Documentation: https://docs.posit.co/connect/admin/user-management/

FAIL **Users can be listed** Connect

src/vip_tests/connect/test_users.py::test_list_users 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› When I list all users
› Then the user list is not empty
› And the test user exists in the user list
```

test_list_users: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017d390>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=13,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_list_users>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
```

```
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The test user named by [auth] username in vip.toml (or VIP_TEST_USERNAME) does not exist on Connect
- Connect has no users other than the bootstrap administrator

Suggested next steps:
1. Create the test user in Connect's admin panel or via `POST /__api__/v1/users`
2. Confirm [auth] username in vip.toml matches an existing Connect username exactly

Documentation: https://docs.posit.co/connect/admin/user-management/

FAIL **Expected R versions are available on Connect** Connect

src/vip_tests/connect/test_runtime_versions.py::test_r_versions 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› And expected R versions are specified in vip.toml
› When I query Connect for available R versions
› Then all expected R versions are present
```

test_r_versions: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40bc40>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_r_versions>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The R version was never installed on the Connect server
- vip.toml's [runtimes] r_versions list is out of date with what was actually installed

Suggested next steps:
1. List installed R versions: `docker exec <container> ls /opt/R` or check the Connect admin panel
2. Install the missing R version, or correct r_versions under [runtimes] in vip.toml

Documentation: https://docs.posit.co/connect/admin/r/

FAIL **Expected Python versions are available on Connect** Connect

src/vip_tests/connect/test_runtime_versions.py::test_python_versions 0.00s

Test procedure

```
› Given Connect is accessible at the configured URL
› And expected Python versions are specified in vip.toml
› When I query Connect for available Python versions
› Then all expected Python versions are present
```

test_python_versions: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017d7f0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=14,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_python_versions>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The Python version was never installed on the Connect server
- vip.toml's [runtimes] python_versions list is out of date with what was actually installed

Suggested next steps:
1. List installed Python versions: `docker exec <container> ls /opt/python` or check the Connect admin panel
2. Install the missing Python version, or correct python_versions under [runtimes] in vip.toml

Documentation: https://docs.posit.co/connect/admin/python/

FAIL **Expected Quarto versions are available** Connect

src/vip_tests/connect/test_runtime_versions.py::test_quarto_versions 0.00s

Test procedure

› Given Connect is accessible at the configured URL
› When I query Connect for available Quarto versions
› Then at least one Quarto version is available

test_quarto_versions: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c356c80>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=20,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_quarto_versions>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Quarto was never installed on the Connect server
- Quarto is installed but not on Connect's configured search path

Suggested next steps:

1. Check installed Quarto versions in the Connect admin panel under Versions
2. Install Quarto and confirm Connect's Quarto.Executable / search path configuration

Documentation: https://docs.posit.co/connect/admin/quarto/

FAIL **Connect system checks can be run and the report downloaded** Connect

src/vip_tests/connect/test_system_checks.py::test_connect_system_checks 0.00s

Test procedure

› Given Connect is accessible at the configured URL
› And a valid API key is configured
› When I trigger a new system check run via the Connect API
› Then the system check report is returned
› And I can download the system check report artifact

test_connect_system_checks: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017e3c0>

    def __enter__(self):
```

```
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_connect_system_checks>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- The API key does not have administrator privileges
- A system check itself is failing on the host (e.g. missing sandbox binary, disk space)
- The system check run timed out before reaching a 'done' status

Suggested next steps:
1. Re-run system checks from the Connect admin panel under System Information
2. Review the individual check results for the specific failing check
3. Verify the API key belongs to an administrator

Documentation: https://docs.posit.co/connect/user/system-checks/

Package Manager

5 tests — 5 failed

FAIL **CRAN mirror is accessible** Package Manager

`src/vip_tests/package_manager/test_repos.py::test_cran_mirror  0.00s`

Test procedure

```
› Given Package Manager is running
› When I query the CRAN repository for the "Matrix" package
› Then the package is found in the repository
```

test_cran_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f816b58d0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^
```

```
/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_cran_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No CRAN-type repository is configured in Package Manager
- The CRAN source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **Bioconductor mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_bioconductor_mirror 0.00s

Test procedure
› Given Package Manager is running
› When I query the Bioconductor repository for the "BiocGenerics" package
› Then the package is found in the repository

test_bioconductor_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c4da430>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=18, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
```

```
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_bioconductor_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No Bioconductor-type repository is configured in Package Manager
- The Bioconductor source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **OpenVSX mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_openvsx_mirror 0.00s

Test procedure

```
› Given Package Manager is running
› When I query the OpenVSX repository for the "golang.Go" extension
› Then the package is found in the repository
```

test_openvsx_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c4da350>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=23, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_openvsx_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?
```

```
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No OpenVSX-type repository is configured in Package Manager
- The OpenVSX source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **PyPI mirror is accessible** Package Manager

src/vip_tests/package_manager/test_repos.py::test_pypi_mirror 0.00s

Test procedure

```
› Given Package Manager is running
› When I query the PyPI repository for the "requests" package
› Then the package is found in the repository
```

test_pypi_mirror: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8166c440>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=13, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_pypi_mirror>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- No PyPI-type repository is configured in Package Manager
- The PyPI source has not finished its initial sync

Suggested next steps:
1. Check configured repositories: `rspm list repos` on the Package Manager host
2. Check sync status: `rspm list sources` and wait for the initial sync to complete

Documentation: https://docs.posit.co/rspm/admin/repositories/

FAIL **At least one repository is configured** Package Manager

src/vip_tests/package_manager/test_repos.py::test_repo_exists 0.00s

Test procedure

› Given Package Manager is running
› When I list all repositories
› Then at least one repository exists

test_repo_exists: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027d8b3ee0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is running', line_number=28, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_repo_exists>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Package Manager was installed but no repositories were ever created
- Repositories exist but were deleted or renamed

Suggested next steps:

1. Create a repository: `rspm create repo --name=<name> --type=<cran|python|bioconductor>`
2. List existing repositories: `rspm list repos`

Documentation: https://docs.posit.co/rspm/admin/repositories/

Prerequisites

6 tests — 6 failed

FAIL **Workbench server is reachable** Prerequisites

src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Workbench] 0.00s

Test procedure

› Given <product> is configured in vip.toml
› When I request the <product> health endpoint
› Then the server responds with a successful status code

test_product_server_is_reachable[Workbench]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017c6e0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Workbench is configured in vip.toml', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_product_server_is_reachable[Workbench]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

FAIL **Package Manager server is reachable** Prerequisites

src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Package Manager] 0.00s

Test procedure

› Given <product> is configured in vip.toml
› When I request the <product> health endpoint
› Then the server responds with a successful status code

test_product_server_is_reachable[Package Manager]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017f310>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is configured in vip.toml', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
```

```
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_product_server_is_reachable[Package Manager]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

FAIL **This check intentionally fails to demonstrate failure rendering** Prerequisites

As someone evaluating a VIP report

`src/vip_tests/prerequisites/test_expected_failure.py::test_intentional_failure  0.00s`

Test procedure

› Given a check that is written to fail on purpose
› When the check runs as part of this example report
› Then it fails by design, not because anything is actually broken

test_intentional_failure: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017fa10>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='a check that is written to fail on purpose', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_intentional_failure>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- This check is designed to always fail, so the example report has a failure card to show
- No product or configuration problem is involved

Suggested next steps:
1. No action needed. This failure is intentional and does not indicate a real problem
2. It only runs when VIP_ENABLE_EXPECTED_FAILURE_DEMO=1 is set

Documentation: https://github.com/posit-dev/vip/issues/73

FAIL **Connect version matches configuration** Prerequisites

src/vip_tests/prerequisites/test_versions.py::test_connect_version 0.00s

Test procedure

```
› Given Connect is configured in vip.toml with a version expectation
› When I fetch the Connect server version
› Then the Connect version matches the configured value
```

test_connect_version: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017dc50>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml with a version expectation',
    line_number=8, indent=4, keyword='Given', docstring=None, datatable=None, failed=False,
    scenario=None, background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_connect_version>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- Connect was upgraded but vip.toml's [connect] version was not updated
- vip.toml points at the wrong Connect deployment

Suggested next steps:
1. Check the running version: `curl -H 'Authorization: Key <key>' <connect-url>/__api__/server_settings`
2. Update the version field under [connect] in vip.toml to match

Documentation: https://docs.posit.co/connect/admin/server-management/

FAIL **Package Manager version matches configuration** Prerequisites

src/vip_tests/prerequisites/test_versions.py::test_package_manager_version 0.00s

Test procedure

```
› Given Package Manager is configured in vip.toml with a version expectation
› When I fetch the Package Manager server version
› Then the Package Manager version matches the configured value
```

test_package_manager_version: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8017fe70>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Package Manager is configured in vip.toml with a version
    expectation', line_number=18, indent=4, keyword='Given', docstring=None, datatable=None,
    failed=False, scenario=None, background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_package_manager_version>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Package Manager was upgraded but vip.toml's [package_manager] version was not updated
- vip.toml points at the wrong Package Manager deployment

Suggested next steps:

1. Check the running version: `curl <pm-url>/__api__/status`
2. Update the version field under [package_manager] in vip.toml to match

Documentation: https://docs.posit.co/rspm/admin/upgrading/

FAIL **Connect server is reachable** Prerequisites

src/vip_tests/prerequisites/test_components.py::test_product_server_is_reachable[Connect] 0.00s

Test procedure

```
› Given <product> is configured in vip.toml
› When I request the <product> health endpoint
› Then the server responds with a successful status code
```

test_product_server_is_reachable[Connect]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b000>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
```

```
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_product_server_is_reachable[Connect]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Security

3 tests — 3 failed

FAIL **Invalid API key returns 401** Security

src/vip_tests/security/test_error_handling.py::test_invalid_api_key_returns_401 0.00s

Test procedure

› Given Connect is configured in vip.toml
› When I make an API request to Connect with an invalid key
› Then the response status is 401

test_invalid_api_key_returns_401: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f805b3930>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=13, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_invalid_api_key_returns_401>
```

```
    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- An upstream proxy or load balancer is granting access before Connect validates the key
- Connect's authentication is misconfigured to accept any Authorization header

Suggested next steps:
1. Reproduce with `curl -i -H 'Authorization: Key invalid' <connect-url>/__api__/v1/content`
2. Check any reverse proxy or gateway in front of Connect for auth bypass rules

Documentation: https://docs.posit.co/connect/admin/authentication/

FAIL **Unauthenticated API request returns 401** Security

src/vip_tests/security/test_error_handling.py::test_unauthenticated_api_request_returns_401
0.00s

Test procedure

```
› Given Connect is configured in vip.toml
› When I make an unauthenticated API request to Connect
› Then the response status is 401
```

test_unauthenticated_api_request_returns_401: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c356f20>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=8, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_unauthenticated_api_request_returns_401>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- An upstream proxy or load balancer is granting access before Connect sees the request
- Connect's authentication is misconfigured to allow anonymous access to authenticated endpoints

Suggested next steps:
1. Reproduce with `curl -i <connect-url>/__api__/v1/content` and inspect the status code
2. Check any reverse proxy or gateway in front of Connect for auth bypass rules

Documentation: https://docs.posit.co/connect/admin/authentication/

FAIL **Non-existent endpoint returns 404** Security

src/vip_tests/security/test_error_handling.py::test_nonexistent_endpoint_returns_404 0.00s

Test procedure

```
› Given Connect is configured in vip.toml
› When I request a non-existent endpoint on Connect
› Then the response status is 404
```

test_nonexistent_endpoint_returns_404: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c40b3f0>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Connect is configured in vip.toml', line_number=18, indent=4,
    keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_nonexistent_endpoint_returns_404>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:
- An upstream proxy or load balancer is intercepting the request before Connect replies
- A reverse proxy is rewriting unmatched paths to a different response

Suggested next steps:
1. Reproduce with `curl -i <connect-url>/__api__/v1/does-not-exist-vip-test`
2. Check reverse proxy path-rewrite and catch-all rules in front of Connect

Documentation: https://docs.posit.co/connect/admin/proxy/

Workbench

2 tests — 2 failed

FAIL **User can sign out of Workbench** Workbench

`src/vip_tests/workbench/test_auth.py::test_workbench_signout[chromium]  0.00s`

Test procedure

```
› Given Workbench is accessible and I am logged in
› When I sign out of Workbench
› Then I am redirected to the Workbench login page
```

test_workbench_signout[chromium]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x771f8031f930>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Workbench is accessible and I am logged in', line_number=14,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x771f8177ae40>
node = <Function test_workbench_signout[chromium]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- The sign-out button or menu changed location between Workbench versions
- A reverse proxy is caching the post-logout redirect

Suggested next steps:

1. Manually sign out in a browser and confirm you land back on the login page
2. Check any reverse proxy or CDN in front of Workbench for cached redirects

Documentation: https://docs.posit.co/ide/server-pro/authenticating_users/authenticating_users.html

FAIL **User can log in to Workbench via the web UI** Workbench

`src/vip_tests/workbench/test_auth.py::test_workbench_login[chromium]  0.00s`

Test procedure

```
› Given Workbench is accessible at the configured URL
› When a user navigates to the Workbench login page and enters valid credentials
› Then the Workbench homepage is displayed
› And the current user element is visible and non-empty in the header
```

test_workbench_login[chromium]: an unexpected error occurred: AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you mean: '_arg2node2fixturedefs'?

```
self = <contextlib._GeneratorContextManager object at 0x7a027c4da270>

    def __enter__(self):
        # do not keep args and kwds alive unnecessarily
        # they are only needed for recreation, which is not possible anymore
        del self.args, self.kwds, self.func
        try:
>           return next(self.gen)
                   ^^^^^^^^^^^^^^

/usr/lib/python3.14/contextlib.py:141:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:136: in inject_fixturedefs_for_step
    fixturedefs = list(find_fixturedefs_for_step(step=step, fixturemanager=fixturemanager,
        node=node))
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

step = Step(type='given', name='Workbench is accessible at the configured URL', line_number=8,
    indent=4, keyword='Given', docstring=None, datatable=None, failed=False, scenario=None,
    background=None)
fixturemanager = <_pytest.fixtures.FixtureManager object at 0x7a027d94ee40>
node = <Function test_workbench_login[chromium]>

    def find_fixturedefs_for_step(step: Step, fixturemanager: FixtureManager, node: Node) ->
        Iterable[FixtureDef[Any]]:
        """Find the fixture defs that can parse a step."""
        # happens to be that _arg2fixturedefs is changed during the iteration so we use a copy
>       fixture_def_by_name = list(fixturemanager._arg2fixturedefs.items())
                                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AttributeError: 'FixtureManager' object has no attribute '_arg2fixturedefs'. Did you
    mean: '_arg2node2fixturedefs'?

.venv/lib/python3.14/site-packages/pytest_bdd/scenario.py:56: AttributeError
```

Likely causes:

- Incorrect test credentials in vip.toml or VIP_TEST_USERNAME/VIP_TEST_PASSWORD
- Auth provider misconfiguration (SAML, OIDC, LDAP, PAM)
- Workbench login page markup changed or is unreachable

Suggested next steps:

1. Verify credentials work by logging in manually at the Workbench URL
2. Check the auth provider configuration in rserver.conf / auth-pam or auth-saml settings
3. If using an external IdP, try `--interactive-auth` or `--headless-auth` instead of a password test

Documentation: https://docs.posit.co/ide/server-pro/authenticating_users/authenticating_users.html